Exp No:18 Experiment to understand Matplotlib library use cases in Data Science through visualization

CODE

```
import pandas as pd

import numpy as np

import seaborn as sns

import matplotlib.pyplot as plt

%matplotlib inline

data=pd.read_csv('/content/Iris_Dataset.csv')

data={ { ‘Id’ = (1,2,3,4,5) , ‘SepalLengthCm= (5.1, 4.9, 4.7, 4.6, 5.0),

        ‘SepalWidthCm= (3.5, 3.0, 3.2, 3.1, 3.6),

        ‘PetalLengthCm= (1.4, 1.4, 1.3, 1.5, 1.4),

        ‘PetalWidthCm = (5.1, 4.9, 4.7, 4.6, 5.0),

}

data.info()

data.describe()

data.value_counts('variety')

sns.countplot(x='variety',data=data,)

plt.show()

dummies=pd.get_dummies(data.variety)

FinalDataset=pd.concat([pd.get_dummies(data.variety),data.iloc[:,[0,1,2,3]]],axis=1)

FinalDataset.head()
```

```
sns.scatterplot(x='SepalLengthCm',y='SepalWidthCm',hue='variety',data=data,)

sns.scatterplot(x='PetalLengthCm',y='PetalWidthCm',hue='variety',data=data,)

sns.pairplot(data,hue='variety',height=3);

plt.show()

sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalLengthCm').add_legend();

plt.show();

sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalWidthCm').add_legend();

plt.show();

sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalLengthCm').add_legend();

plt.show();

sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalWidthCm').add_legend();

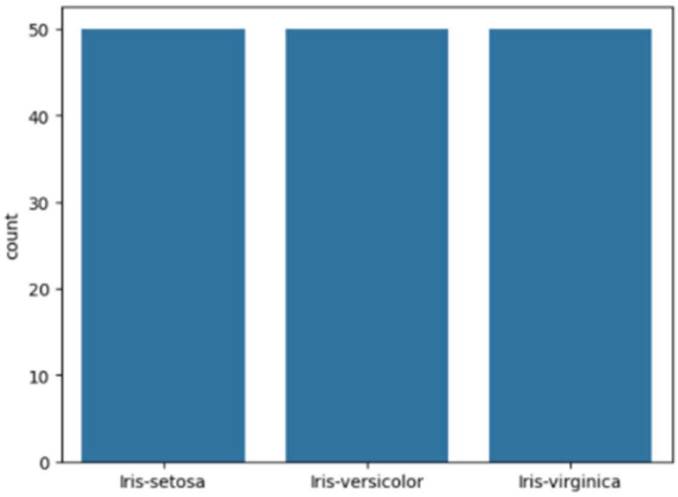
plt.show();
```

OUTPUT

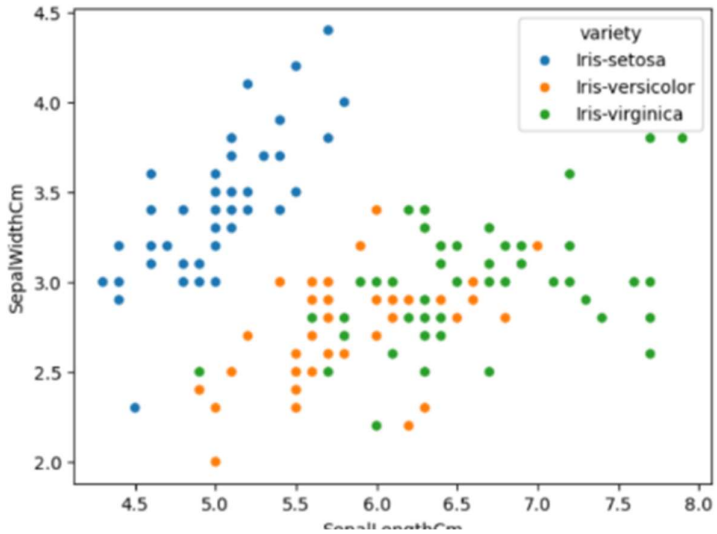
	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	variety
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

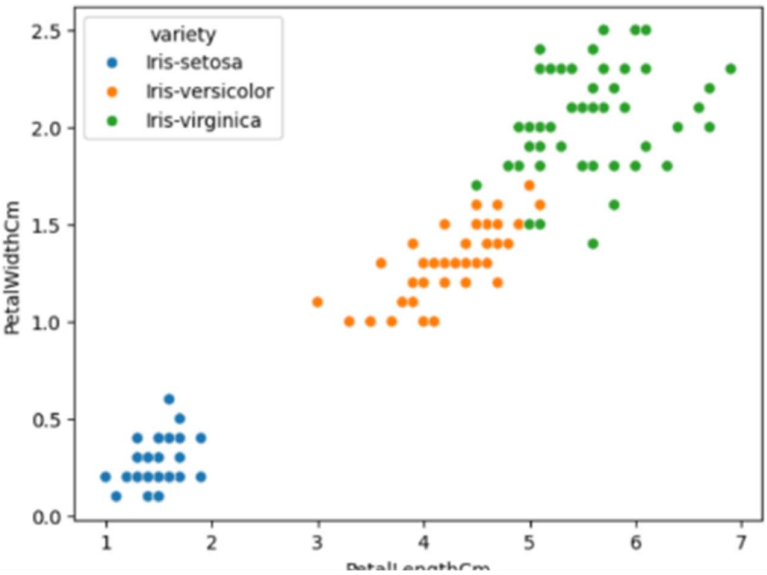
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   variety        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

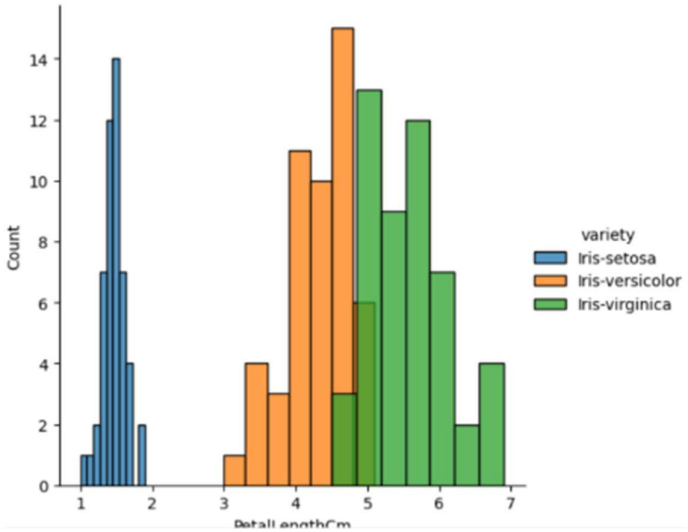
	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000

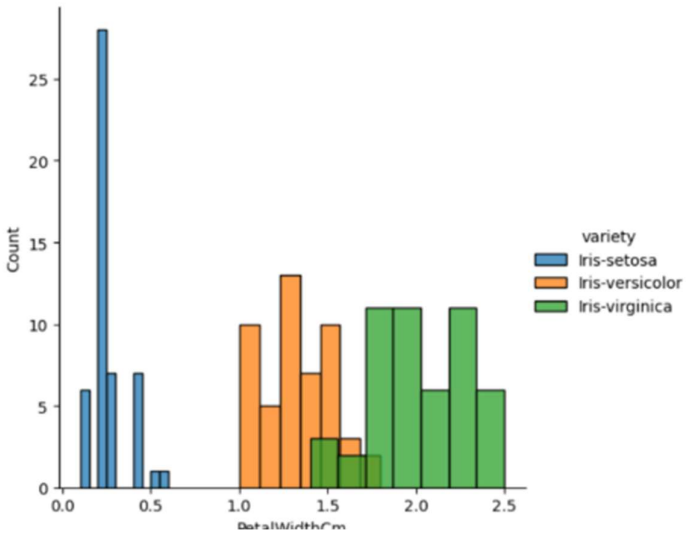


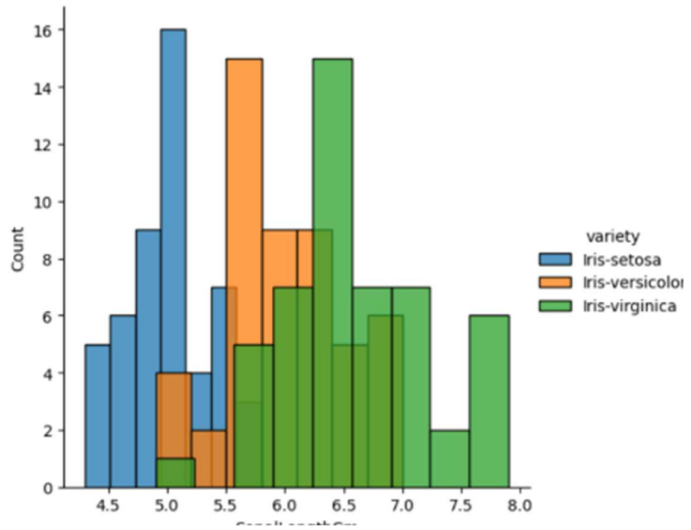
	Iris-setosa	Iris-versicolor	Iris-virginica	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm
0	True	False	False	1	5.1	3.5	1.4
1	True	False	False	2	4.9	3.0	1.4
2	True	False	False	3	4.7	3.2	1.3
3	True	False	False	4	4.6	3.1	1.5

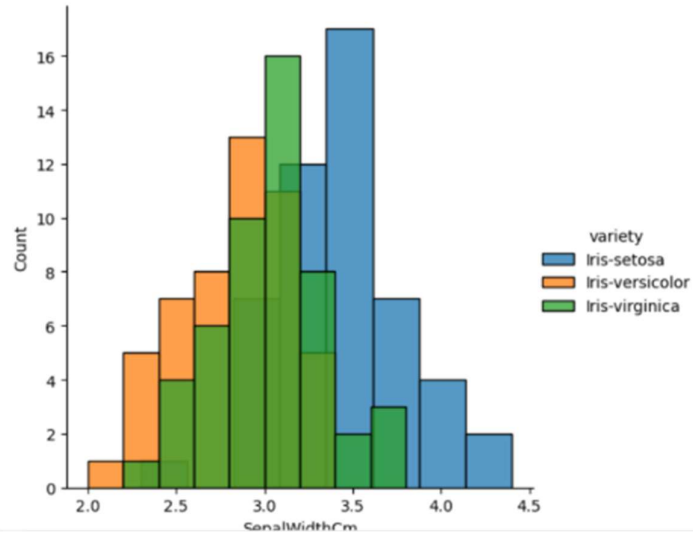

variety
Iris-setosa
Iris-versicolor
Iris-virginica
SepalWidthCm
4.5
4.0
3.5
3.0
2.5
2.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0

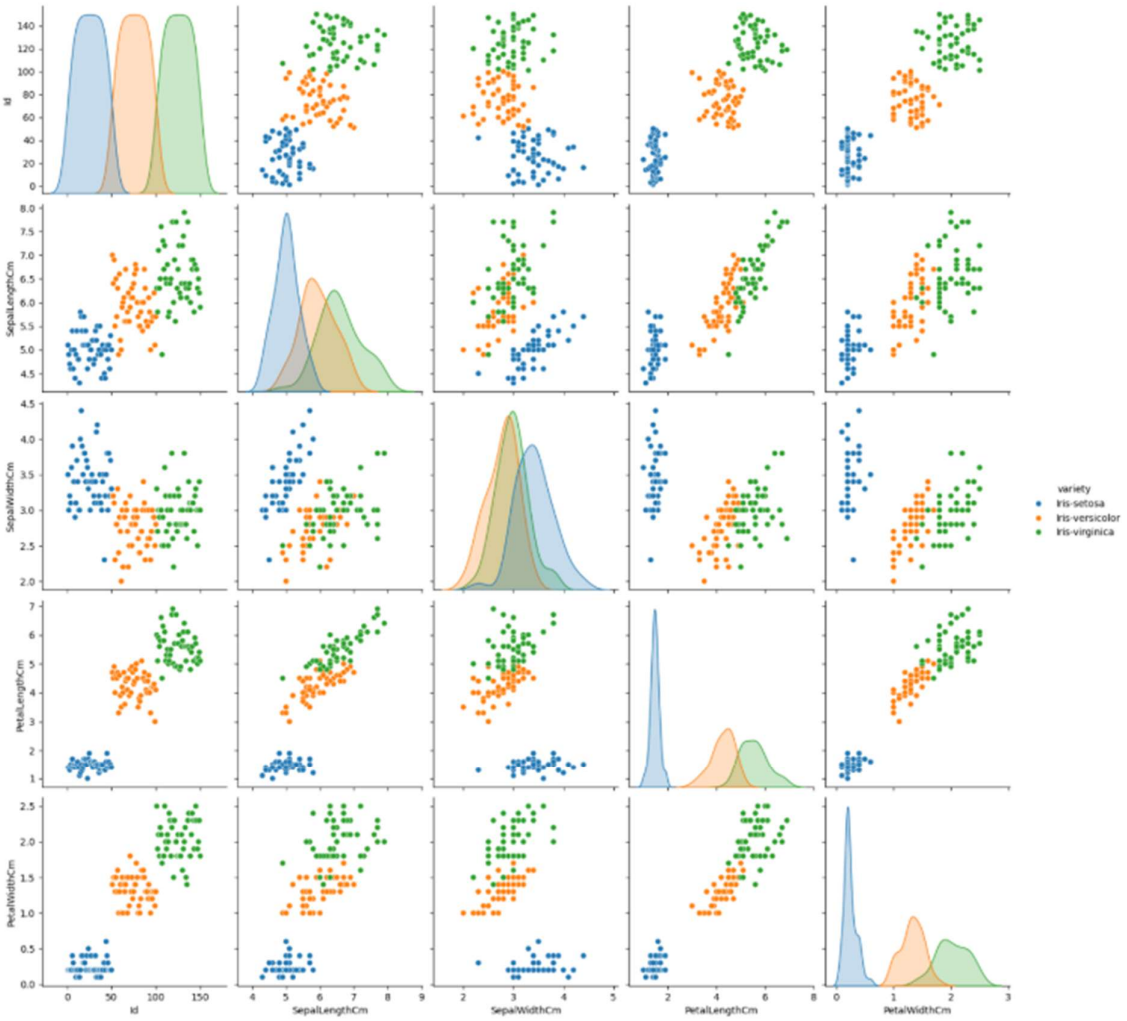

variety
Iris-setosa
Iris-versicolor
Iris-virginica
PetalWidthCm
2.5
2.0
1.5
1.0
0.5
0.0
1
2
3
4
5
6
7


Count
14
12
10
8
6
4
2
0
1
2
3
4
5
6
7
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
25
20
15
10
5
0
0.0
0.5
1.0
1.5
2.0
2.5
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
16
14
12
10
8
6
4
2
0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
16
14
12
10
8
6
4
2
0
2.0
2.5
3.0
3.5
4.0
4.5
variety
Iris-setosa
Iris-versicolor
Iris-virginica



RESULT: Hence understand Matplotlib library using cases in Data Science through visualization has been studied.